



Autonomous Vehicle Platoon Packet Based Control Problem Under Denial-of-Service Attacks

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Abstract. The present work tackles the $\mathcal{H}_\infty$ Packet-based control design in the autonomous vehicle platoon (AVP) problem based on digital redesign of a linear parameter-varying system under the assumption of the presence of Denial of Service (DoS) attacks. The main novelty in this paper is the parameter-dependent modeling and the added assumption of DoS attacks during the control design process. This assumption is particularly relevant in the AVP problem since all the vehicles exchange data using an autonomous vehicle network (AVN), and those networks can be susceptible to attack from malicious agents. To illustrate the viability of the proposed approach some simulations are provided considering the time-invariant model of the AVP, and the parameter-varying model with and without the presence of DoS attacks.

Keywords: DoS attacks · Cyber physical systems · $\mathcal{H}_\infty$ guaranteed cost · Platoon of vehicles

1 Introduction

In the last decade, the discussion about self-driving vehicles has gained traction in academia. The main objective of designing a fully automated vehicle is to remove human error. In [1, 2] the authors made projections that the usage of autonomous vehicles (AV) would be worldwide spread by 2040. Among the different issues that permeate the class of autonomous vehicles one that stands out is the platoon problem, also known as Autonomous vehicle platoon (AVP), which is one of the most realistic and closer to being a reality. Several practical trials have been made [3, 4].

One of the aspects of the implementation of AVP is the usage of Autonomous Vehicular Networks (AVN) [5, 6]. Since it is essential to connect the AVP in an AVN, some new challenges regarding the network must be taken into consideration. Among these challenges, one may cite the presence of malicious attacks against the AVP [7].

There are some examples of works in the literature that tackle the AVP control problem such as [8] AVP control approach that uses real-time MPC-based control design. In [9] the AVP problem is modeled as a multi-agent system assuming that all vehicles will share some internal information, which the authors named as Inter-Vehicular Communication (IVC) systems. The results in [10] solve the vehicle platoon problem using cooperative adaptive cruise control, assuming that the communication between vehicles is made via an idealized fixed information flow topology, which assumes that the communication is guaranteed. [11] provides an Adaptive Cruise Control design where the platoon is modeled using a hierarchical structure and the controller is designed using Linear Quadratic Regulators (LQR). [12] uses Cyber-Physical Systems (CPS) to model the platoon and provide the comparison between two types of control designs LQR and Generalized Predictive Control (GPC). None of the aforementioned examples consider the presence of malicious agents in the AVP. Moreover, the model implemented in all those examples is the longitudinal vehicle dynamics Linear Time-Invariant (LTI) model in the continuous-time domain.

From the above discussion, two major aspects that should be considered during the control design process for an AVP are *i*) as in any model-based control problem the model must be as precise as possible, and *ii*) the eventual presence of malicious attacks in the network must be taken into account. As far as the authors are aware, there are no works in the literature that tackle the AVP problem under the cyber-physical framework with time-varying parameters while also considering malicious attacks in the design process.

In the search for a solution to the problem of AVP under malicious attacks in the communication channels, we propose the following methodology. Based on the lateral dynamic car model [13], a parameter-dependent state-space model is obtained, which allows incorporating uncertain or time-varying dynamics, a more realistic approach for AVP problems when using wireless communication channels for malicious attacks. The parameter-dependent AVP in discrete-time based on the continuous-time car model uses the approach called *digital redesign* [14], a method that guarantees the state convergence in the discrete-time and continuous-time models.

Analogous to [15, 16], the strategy should include attacks from malicious agents in its design, more specifically Denial of Service (DoS) attacks. The approach from [15] is used as it considers exogenous disturbances and $\mathcal{H}_\infty$ cost in a packet-based state feedback strategy.

The major contributions of this paper are: The design of a robust Packet-based state feedback controller for discrete-time linear parameter-varying (LPV) systems that assumes that the system is subjected to exogenous disturbance and DoS attacks. Since the objective is to control the lateral dynamics of each vehicle in the platoon, the mathematical model implemented is the bicycle model [17–20], which is a well-established model in the literature, allowing to link the state-of-the-art in DoS attacks to the real AVP problem. Different simulations are provided to validate the proposed approach, which are compared with a classic strategy from the literature.

This paper is organized as follows. The Sect. 2 describes the Vehicle Platoon problem, Sect. 3 provides notation and the mathematical foundation, Sect. 4 describes the main contribution, Sect. 5 presents the simulation results and the discussions, and in Sect. 6 some final remarks are made.

Notation: The general notation considered throughout the paper is standard. The set of natural numbers is denoted by $\mathbb{N}$ and the n -th dimensional Euclidean space with the usual norm $\|\cdot\|$ is expressed by $\mathbb{R}^n$. The set of real matrices with dimension $n \times m$ is denoted by $\mathbb{R}^{n \times m}$. The symbol $'$ indicates the transposition of a matrix or a vector. The symbol $\star$ denotes a block induced by symmetry in a symmetric matrix. Assume that the time-varying matrices have a polytopic parameter dependency. This parameter is modeled in the unit simplex Λ_V , where the dimension V is the number of vertices of the polytope. A generic matrix is given by

$$X(\theta(k)) = \sum_{v=1}^V \theta_v(k) X_v, \quad \theta(k) \in \Lambda_V, \quad (1)$$

where $\sum_{v=1}^V \theta_v = 1$ and $\theta_v \geq 0$, $v \in 1, \dots, V$.

2 Autonomous Vehicle Platoon Modeling

An AVP system is a convoy of vehicles problem, one following another while tracking the same route, as shown in Fig. 1. The first vehicle (the leader) is controlled by an arbitrary input, and the followers are controlled by a control input. The goal is to provide a control law that keeps the followers on the same path as the leader. Assuming that all the vehicles are the same, the state-space dynamic model for the continuous-time augmented system is

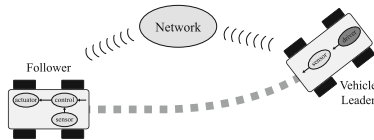


Fig. 1. Scheme of AVP system.

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} A_c(\theta(t)) & 0 \\ 0 & A_c(\theta(t)) \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ B_c(\theta(t)) \end{bmatrix} u(t) + \begin{bmatrix} B_c(\theta(t)) & 0 \\ 0 & e_d B_c(\theta(t)) \end{bmatrix} w(t), \\ z(t) &= \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 0 \\ e_u \end{bmatrix} u(t). \end{aligned} \quad (2)$$

where, $A_c(\theta(t))$ is the matrix with the dynamic of a single vehicle, $B_c(\theta(t))$ is the car input matrix, both dependent on the time-varying parameter. $x(t)$ is the augmented states vector, $u(t)$ represents the control input, and $w(t)$ is an exogenous input. The output $z(t)$ is the variation between both cars' state vectors (leader and follower). Furthermore, the scalar e_d is an influence factor of disturbance on the second input, and the scalar e_u is an error-adjustment constant (more information can be found in [13]).

2.1 Car Model's Lateral Dynamics

The single-car state-space dynamic model ($A_c(\theta(t))$ and $B_c(\theta(t))$) are obtained based on the car's lateral dynamics, a classical model in the literature, presented in Fig. 2 [21, Chapter 2]. For a single car the standard state-space dynamic model is given by

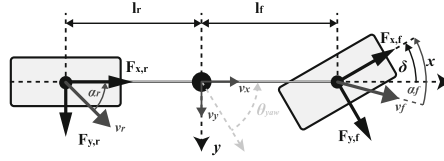


Fig. 2. Free-body diagram of single-track model.

$$\begin{bmatrix} \dot{v}_y(t) \\ \dot{r}(t) \end{bmatrix} = \begin{bmatrix} \frac{-(c_r + c_f)}{mv_x} & \frac{(c_r l_r - c_f l_f)}{mv_x} \\ \frac{(l_r c_r - l_f c_f)}{Jv_x} & \frac{(l_r^2 c_r + l_f^2 c_f)}{Jv_x} \end{bmatrix} \begin{bmatrix} v_y(t) \\ r(t) \end{bmatrix} + \begin{bmatrix} \frac{c_f}{m} \\ \frac{l_f c_f}{J} \end{bmatrix} \delta(t), \quad (3)$$

where m denotes the mass of the vehicle, J represents the inertia in the yaw axis, l_r and l_f are the distance between the center-of-gravity and the extremes of the axles, c_r and c_f are the coefficients of cornering stiffness of the axles, F_y and F_x are the forces in the y -axis and x -axis (for both front and rear axles), v_y and v_x are the vehicle velocities in their respective axes. The steering angle of the front axle is represented by $\delta(t)$, $r(t)$, being a variation on the yaw angle, and finally, α denotes the slip angle on both axles. Hence, the LTI system is given by (3).

2.2 Parameter-Dependent AVP Modeling

In this section, the interval for tangential velocity in the x -axis is represented by $v_x(t) \in [v_{x,\min}, v_{x,\max}]$, which is a bounded time-varying parameter. According to [22], the system in (3) can be written in polytopic form based on the unit simplex (1). Considering the new bounded time-varying parameters; $\varepsilon(\theta(t)) = \frac{1}{v_x(t)} \in [\underline{\varepsilon}, \bar{\varepsilon}]$ and $\zeta(\theta(t)) = v_x(t) \in [\underline{\zeta}, \bar{\zeta}]$, which allow us to rewrite the system as an affine linear parameter-dependent system, and, using (1) we can get (3) as the following state-space model,

$$\dot{x}(t) = \hat{A}(\theta(t))x(t) + \hat{B}(\theta(t))u(t), \quad (4)$$

Both $\hat{A}(\theta(t))$ and $\hat{B}(\theta(t)) \in \Lambda_4$, a simplex with four vertices in the form (1), describing a combination of the limits of the intervals, as given by

$$\begin{aligned} A_1 &= \begin{bmatrix} a_{11}\underline{\varepsilon} & a_{12}\underline{\varepsilon} - \underline{\zeta} \\ a_{21}\underline{\varepsilon} & a_{22}\underline{\varepsilon} \end{bmatrix}; A_2 = \begin{bmatrix} a_{11}\bar{\varepsilon} & a_{12}\bar{\varepsilon} - \bar{\zeta} \\ a_{21}\bar{\varepsilon} & a_{22}\bar{\varepsilon} \end{bmatrix} \\ A_3 &= \begin{bmatrix} a_{11}\underline{\varepsilon} & a_{12}\underline{\varepsilon} - \bar{\zeta} \\ a_{21}\underline{\varepsilon} & a_{22}\underline{\varepsilon} \end{bmatrix}; A_4 = \begin{bmatrix} a_{11}\bar{\varepsilon} & a_{12}\bar{\varepsilon} - \bar{\zeta} \\ a_{21}\bar{\varepsilon} & a_{22}\bar{\varepsilon} \end{bmatrix}, \end{aligned} \quad (5)$$

$$\begin{aligned}
a_{11} &= \frac{-(c_r + c_f)}{m}, & a_{12} &= \frac{(c_r l_r - c_f l_f)}{m}, \\
a_{21} &= \frac{(l_r c_r - l_f c_f)}{J}, & a_{22} &= \frac{(l_r^2 c_r + l_f^2 c_f)}{J}.
\end{aligned} \tag{6}$$

In this scenario, $\hat{B}(\theta(t))$ does not depend on the time-varying parameter. Note that in this setup there are some levels of conservatism, since the new variables $\varepsilon(k)$ and $\zeta(k)$ are nonlinearly dependent on $v_x(k)$, but are modeled as independent.

2.3 Discrete-Time Parameter-Dependent Model for Platoon Car

Both (2) and (3) are presented in continuous-time, however, due to the Cyber-Physical System design, it is necessary to use the discrete-time APV model. To ensure state convergence of the actual continuous-time APV using its discrete-time representation convergence, a technique which consists of a Taylor series expansion on uncertain models presented in [14] was utilized, as described below

$$\begin{aligned}
A_l(\theta) &= \sum_{j=0}^l \frac{A_{cont}(\theta)^j}{j!} T^j, & B_l(\theta) &= \sum_{j=1}^l \frac{A_{cont}(\theta)^{j-1}}{j!} T^j B_{cont}(\theta), \\
\Delta A_l(\theta) &= e^{A_{cont}(\theta)T} - A_l(\theta), & \Delta B_l(\theta) &= \left(\int_0^T e^{A_{cont}(\theta)s} ds \right) B_{cont}(\theta) - B_l(\theta),
\end{aligned} \tag{7}$$

where $A_l(\theta)$ as well as $B_l(\theta)$ are given by the Taylor series expansion and $\Delta A_l(\theta)$ together with $\Delta B_l(\theta)$ are the residual of the series. The values of sampling period T and Taylor series grade l are chosen to consider the values of $\Delta A_l(\theta)$ and $\Delta B_l(\theta)$. Given the numerical values of the considered system and the sampling period, in the present work, the value of the approximation error is neglected. Consequently, the discrete-time APV model with $v_x(t)$ as a bounded time-varying parameter is expressed by the augmented system given by

$$\begin{aligned}
x(k+1) &= \begin{bmatrix} A_l(\theta(k)) & 0 \\ 0 & A_l(\theta(k)) \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ B_l(\theta(k)) \end{bmatrix} u(k) + \begin{bmatrix} B_l(\theta(k)) & 0 \\ 0 & e_d B_l(\theta(k)) \end{bmatrix} w(k), \\
z(k) &= \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} x(k) + \begin{bmatrix} 0 \\ 0 \\ e_u \end{bmatrix} u(k),
\end{aligned} \tag{8}$$

where the augmented state vector is $x(k) = [v_{y1}(k) \ r_1(k) \ v_{y2}(k) \ r_2(k)]$, the input vector is $u(k) = \delta_2(k)$ and the exogenous input vector is $w(k) = [\delta_1(k) \ p(k)]$, where $\delta_1(k)$ is the arbitrary input of the first vehicle and $p(k)$ is a disturbance in the second vehicle angle of input.

3 Control Design Framework

Consider the following LPV system in the discrete-time domain

$$\mathcal{G} : \begin{cases} x(k+1) = A(\theta(k))x(k) + B(\theta(k))u(k) + E(\theta(k))w(k) \\ z(k) = C_z(\theta(k))x(k) + D_z(\theta(k))u(k) + E_z(\theta(k))w(k) \end{cases}, \tag{9}$$

where $x(k) \in \mathbb{R}^{n_x}$, $u(k) \in \mathbb{R}^{n_u}$, $w(k) \in \mathbb{R}^{n_w}$ and $z(k) \in \mathbb{R}^{n_z}$, respectively denote the state, control input, external disturbance and output signal vectors. Note that the system (9) depends on the parameter $\theta(k)$, which represents the linear parameter-varying behavior of the system.

We can assume that the attacker is energetically bounded, limiting the maximum number of consecutive attacks to N time instants. This assumption is reasonable and found in other works in the literature [15, 16, 23].

By employing the system dynamics under attack, and the robust packet-based state-feedback control law that admits DoS Attacks proposed by [15]

$$u(k_k + s) = K_s x(k_k), \quad (10)$$

where k_k is the time instant when the packet gets to the actuator and depends on the switching point $s = 0, 1, \dots, \sigma(k_k)$, $\sigma(k_k)$ is a time-varying switching signal which takes values in a finite set $M \triangleq \{0, 1, \dots, N\}$, representing the number of consecutive DoS attacks initiated at the time $k_k + 1$, with a maximum of N attacks. If $\sigma(k_k) = 0$, there are no DoS attacks at the time $k_k + 1$.

When the system is under attack, the relation between two switching points is described as follows

$$k_{k+1} = k_k + \sigma(k_k) + 1, \quad (k_0 = 0). \quad (11)$$

Moreover, $U(k_k) = [u(k_k)', u(k_k + 1)', \dots, u(k_k + N)']'$ is the packet with $N + 1$ signals that is transmitted every time instant the system is not under attack. With the arrival of a new packet, the proposed control strategy is implemented under the following rules:

- i) If a new packet gets to the actuator at the time $k_k + 1$, $u(k_k)$ is the only utilized input from $U(k_k)$.
- ii) If at $k_k + 1$ a DoS attack is initiated, the previously designed control inputs $u(k_k + 1), \dots, u(k_k + s)$ will be applied successively until a new packet arrives.

Inspired by the formulation in [15], N consecutive DoS Attacks are considered. The following result seeks to design robust packet-based state-feedback gains with an upper bound γ for the closed-loop $\mathcal{H}_\infty$ performance of the system.

Theorem 1. *If there exist the matrices $P_i(\theta_{k_k}) = P_i(\theta_{k_k})' > 0 \in \mathbb{R}^{n_x \times n_x}$, $X \in \mathbb{R}^{n_x \times n_x}$, $Z_i \in \mathbb{R}^{n_u \times n_x}$, and a scalar γ such that*

$$\begin{bmatrix} A_j & \Phi_i & \mathcal{B}_i & 0_{n_x \times n_y} \\ \star & -P_i(\theta_{k_k}) & 0_{n_x \times (i+1)n_w} & X' C_z(\theta_{k_k})' + Z_i' D_z(\theta_{k_k})' \\ \star & \star & -\gamma^2 I & \mathcal{D}_i' \\ \star & \star & \star & -I \end{bmatrix} < 0, \quad (12)$$

hold for all $(\theta_{k_k}, \theta_{k_k+1}, \dots, \theta_{k_k+N}) \in \Lambda_V \times \Lambda_V \times \dots \times \Lambda_V$ with

$$\Phi_i = \prod_{l=0}^i A(\theta_{k_k+(i-l)})X + \sum_{n=0}^i \Theta_n B(\theta_{k_k+(i-n)})Z_{i-n}, \quad (13)$$

$$\Theta_n = \prod_{r=1}^n A(\theta_{k_k+(i-r+1)}), \quad \Theta_0 = I, \quad (14)$$

$$A_j = P_j(\theta_{k_k+1}) - X - X' \quad (15)$$

where $\mathcal{B}_i$ is

$$\mathcal{B}_i(\theta_{k_k}, \dots, \theta_{k_k+i}) = [\Psi_m \ \Psi_{m-1} \ \dots \ \Psi_0], \ \Psi_0 = E(\theta_{k_k}), \quad (16)$$

$$\Psi_m = \prod_{z=1}^m A(\theta_{k_k+(N-z+1)})E(\theta_{k_k+(N-m)}), \ m = 1, \dots, i, \quad (17)$$

where $\mathcal{D}_i$ is

$$\mathcal{D}_i = \begin{bmatrix} E_z(\theta_{k_k}) & 0_{n_y \times in_w} \end{bmatrix}, \quad \mathcal{D}_0 = E_z(\theta_{k_k}), \quad (18)$$

and $i, j = 0, \dots, N$, then

$$K_i = Z_i X^{-1}, \quad (19)$$

are the state-feedback control gains that assure the closed-loop system (9) (with a maximum of N consecutive DoS Attacks) to be asymptotically stable for all $\theta_{k_k} \in \Lambda_V$ when $w = 0$. Furthermore, when $w \neq 0$, the guaranteed $\mathcal{H}_\infty$ performance of the switched LPV system is bounded by γ . A higher N results in a more conservative norm. The proof can be found in [15].

4 APV Case Study

To illustrate the feasibility of the proposed approaches, we used a two single-car lateral model. For the numerical test of the continuous-time augmented system (3) use the values of Table 1. For the control design, the matrices that compose the discrete-time augmented system are based on the Taylor Series expansion for uncertain models (7). The AVP parameter-dependent discrete-time model is created using the sampling period of $T = 4$ ms, $v_x(\theta(k)) \in [15, 35]$ m/s and the Taylor Series expansion with $l = 2$. The residual of $\Delta A_l(\theta(k))$ and $\Delta B_l(\theta(k))$ is 0.4025×10^{-3} and 2.3×10^{-3} respectively, which is neglected. For simulation, the initial conditions vector is defined as,

Table 1. Numerical values for simulation.

Parameter	Value
Inertia J	1436.24
Distance l_f	1.165
Distance l_r	1.165
Coefficient c_f	155494.663
Coefficient c_r	155494.663
Mass m	1140

$$x(0) = [v_{y1}(0) \ r_1(0) \ v_{y2}(0) \ r_2(0)]' = [0 \ 0 \ -0.3 \ -0.2]', \quad (20)$$

The exogenous input in continuous-time domain employed in the simulations is

$$w(t) = \begin{bmatrix} \delta_1(t) \\ p(t) \end{bmatrix} = \begin{bmatrix} 0.1e^{-0.02t} \cos(0.5t) \\ 0.1e^{-0.02t} \sin(0.5t) \end{bmatrix}. \quad (21)$$

For the tests the bounded variations of $v_x(t)$ are determined by

$$v_x(t) = 25 + 10 \sin\left(\frac{\pi}{50}t\right). \quad (22)$$

4.1 Attack Detection Methodology

An assumption that is imperative for the solution proposed in this work is that the DoS attacks are detectable. This assumption allows us to detect and measure the duration of an attack, permitting the packet's inputs to be applied according to this information. There are plenty of approaches that provide such information, but those approaches depend on the type of the AVN, if the AVN is a Software Defined Network (SDN) [24], or the AVN uses Message Queuing Telemetry Transport (MQTT) protocol [25]. Another way to gather this information is to implement learning processes as described in [26]. Figure 3 presents the form in which attacks are considered for the analysis.

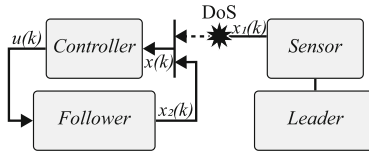


Fig. 3. Scheme of where the DoS attacks affect the system.

5 Results and Discussion

This section presents the performance of the proposed approach based on simulations for the AVP problem. In this instance, we evaluate the precisely known and parameter-dependent systems. The latter is tested respectively without and with the presence of DoS attacks.

Applying Theorem 1 on system (8) described in Sect. 4, the packet-based controller for test corresponds to

$$\begin{bmatrix} K_0 \\ K_1 \\ K_2 \\ K_3 \end{bmatrix} = \begin{bmatrix} -0.1907 & 1.6254 & 0.1930 & -1.6297 \\ -0.6546 & 1.2822 & 0.6522 & -1.2808 \\ -0.4482 & 0.6095 & 0.4479 & -0.6068 \\ -0.1866 & 0.1830 & 0.1870 & -0.1806 \end{bmatrix}. \quad (23)$$

The Packet-based controller was designed considering a maximum of $N = 3$ consecutive attacks and guarantees an upper-bound of the $\mathcal{H}_\infty$ norm with the value of $\gamma = 3.4409$. This N was arbitrarily considered to keep the γ cost low.

Also considering the AVP system under DoS attacks, and for the sake of comparison, a classical discrete-time linear-quadratic regulator¹ was obtained,

$$\mathcal{K} = \begin{bmatrix} 0.5890 & -2.6231 & -0.5890 & 2.6231 \end{bmatrix}, \quad (24)$$

¹ Considering the values of $Q = C_z' C_z$ and $R = 0.001$ (values obtained empirically for the worst operating point of the LTI model $v_{x,35} = 35 \frac{m}{s}$).

and the performance criteria for the different scenarios is given by

$$\mathcal{J} = \frac{1}{T} \int_0^{T=200} x(t)' C_z' C_z x(t), \quad (25)$$

which is the quadratic cost of the continuous-time APV systems output, where subscript c corresponds to controller.

5.1 Time-Invariant Case

Considering a time-invariant speed $v_x(t) = v_x$ (LTI system (3)). The initial condition Eq. (20) and/or exogenous input $w(t)$ by Eq. (21) are employed. Note that, the actual system is in continuous-time, however, the control design is made in the discrete-time domain. Figure 4 shows the state of the leader and follower to LTI case, where it is noted that both controllers offer similar dynamics, obtaining visually unobservable discrepancies.

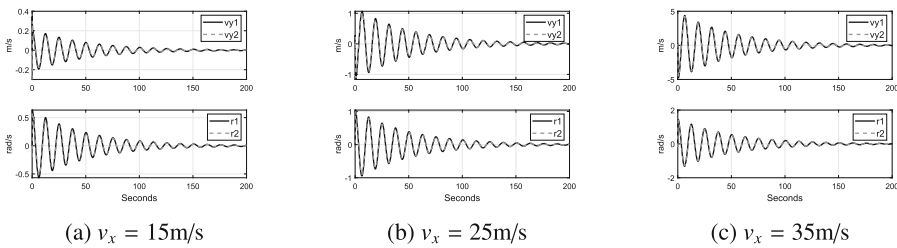


Fig. 4. State of the leader (—) and follower (---) for different tangential velocities v_x .

Table 2 shows the criteria $\mathcal{J}$ based on the control gain obtained by Theorem 1 in this time-invariant case (single operation point).

Table 2. Performance $\mathcal{J}$ for precisely known systems.

Exo. input	Initial condition	$v_x = 15$	$v_x = 25$	$v_x = 35$
0	x_0	0.000020	0.0000021	0.0000197
$w(t)$	0	0.000463	0.000774	0.003096
$w(t)$	x_0	0.000467	0.000778	0.003098

The maximum value of the criteria $\mathcal{J}$ corresponds to 3.098×10^{-3} in $v_x = 35$, consequently the tracking is performed satisfactorily.

Time-Varying Case. Utilizing the LPV system (8), the considered time-varying tangential velocity $v_x(t)$ is described by (22). The $\mathcal{J}$ cost was calculated in two cases: i) x_0 initial conditions and no exogenous inputs; ii) Exogenous inputs of (21) and no initial conditions. Respectively, the costs were 0.021×10^{-3} and 1.865×10^{-3} .

5.2 Time-Varying System Under DoS Attacks

Still taking the LPV system into account, a DoS attack of three consecutive time instants was considered. The interval between attacks is three or one instant. In other words, an attack occurs for three consecutive time instants, then three/one time instants are free of attacks, as seen in Fig. 5. For comparison sake, the classical LQR controller is also tested.

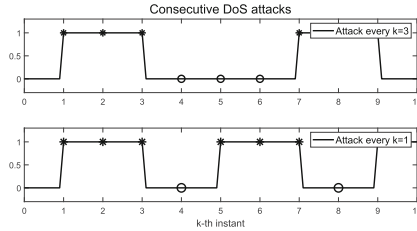
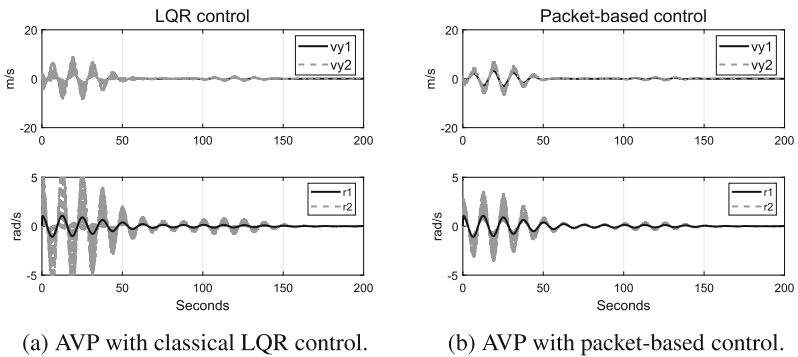


Fig. 5. Sequence of DoS attacks, where (*) is presence of attack and (o) is absence of attack.

The systems under attack lose performance, however, the controller obtained by Theorem 1 tracks the states better than the LQR controller. The system dynamics under DoS attacks can be seen in Fig. 6 and 7. Table 3 reports the performance criteria $\mathcal{J}$.



(a) AVP with classical LQR control.

(b) AVP with packet-based control.

Fig. 6. Time-varying AVP systems under consecutive DoS attacks, the state of the leader (—) and follower (---) for attacks every $k = 3$ instants applied in the control signal.

Table 3. Performance $\mathcal{J}$ under consecutive DoS attacks.

Controller	Every three instants	Every one instant
LQR($v_{x,35}$)	1.67488	12.6320
Theorem 1	0.6947	6.2662

In Fig. 6 there is a higher increase in the maximum magnitude of the follower states using the classical LQR than with the Theorem 1 (Packet-based controller). For example, the value of the second state (r) is approximately 46% more pronounced with the classical gain. Moreover, between 60 and 100 s, the proposed control presents less dispersion. Figure 7 further illustrates this behavior, as the attacks are more frequent.

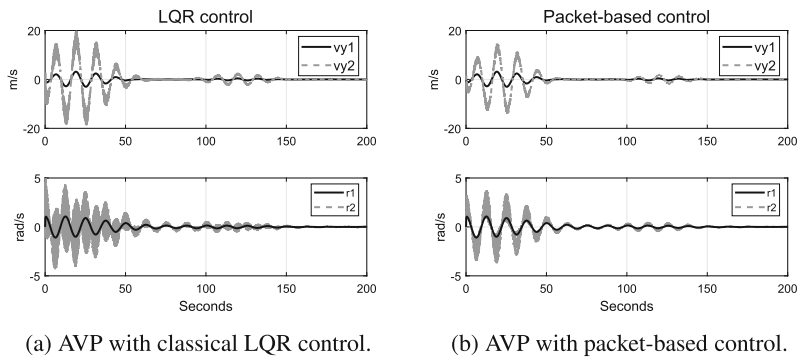


Fig. 7. Time-varying AVP systems under consecutive DoS attacks, the state of the leader (—) and follower (---) for attacks every $k = 1$ instants applied in the control signal.

6 Conclusion

In this paper, the autonomous vehicle platoon control problem is dealt with using a discrete-time design for a robust packet-based state-feedback controller with the presence of DoS attacks based in [15]. The major contribution presented in this work is the addition of the DoS attacks in the controller design for AVP systems. Note that [15] only used arbitrary numeric examples, however, in this paper a discretized continuous-time system was employed. Simulations were conducted comparing a time-invariant model, and a time-varying model with and without attacks, proving the proposed technique's superiority.

In summary, the assumption of eventual malicious DoS attacks is relevant to the controller performance. The next step in this line of research may be the addition of Markovian Jump Linear Systems to model the network behavior, adding the possibility of information loss caused by a packet dropout.

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